

Robotix - Inventory Turnover Analysis

Q1 2024 Report

Author: Michael Rodriguez, CFO

Date: April 15, 2024

Distribution: Executive Team, Board of Directors

Executive Summary

Our Q1 2024 inventory analysis reveals an overall turnover ratio of 3.2x, below the industry benchmark of 3.5-4.0x. Total inventory value stands at \$7.4M, with Days Inventory Outstanding (DIO) at 114.1 days. While showing improvement from Q4 2023 (DIO: 122.3 days), further optimization is needed to reach industry standards.

Key Metrics Overview:

Overall Turnover Ratio: 3.2x

DIO: 114.1 days

Total Inventory Value: \$7.4M

Working Capital Impact: \$2.1M improvement opportunity

Inventory Turnover by Category

Product Category	Turnover Ratio	Avg. Inventory Value	DOI (Days)
Industrial Robots	3.8x	\$2.9M	96.1
Collaborative Robots	2.9x	\$1.8M	125.9
Mobile Robots	2.6x	\$1.4M	140.4
Components & Accessories	4.2x	\$1.3M	86.9
Software Licenses	N/A	N/A	N/A

Detailed Analysis

Raw Materials Inventory

- Total Value: \$2.8M
- Turnover: 4.1x
- Major Components:

- Motors & Actuators: \$980K
- Controllers: \$720K
- Sensors: \$640K
- Frames & Structural: \$460K

Work-in-Progress (WIP)

- Total Value: \$1.9M
- Average Assembly Time: 18 days
- Aging Profile:
 - 0-30 days: 65%
 - 31-60 days: 25%
 - 60 days: 10%

Finished Goods

- Total Value: \$2.7M
- Product Line Distribution:
 - RX-5000 Series: \$1.2M
 - CR-2000 Series: \$800K
 - MR-1000 Series: \$700K

Location Analysis

Facility	Inventory Value	Turnover	Utilization
Minneapolis Mfg	\$4.8M	3.4x	82%
Portland DC	\$2.6M	2.9x	75%

Quarter-over-Quarter Comparison

Q1 2024 vs Q4 2023:

- Overall Turnover: +0.3x improvement
- Total Inventory: -8.2% reduction
- DIO: -8.2 days improvement

Cash Flow Impact

Working Capital Analysis: - Cash tied in inventory: \$7.4M - DSO: 45 days - Cash Conversion Cycle: 159.1 days - Available Working Capital Release: \$2.1M

Risk Assessment

High-Risk Areas: 1. Collaborative Robot Components (120+ days) 2. Legacy Control Systems (180+ days) 3. Custom Sensor Arrays (90+ days)

Recommendations

- 1. **Immediate Actions:** - Implement 15% markdown on CR-2000 legacy models - Accelerate Minneapolis-Portland transfer optimization - Negotiate extended payment terms with key suppliers
- 2. **Strategic Initiatives:** - Deploy advanced demand forecasting system by Q3 - Establish vendor-managed inventory for standard components - Implement cross-docking for fast-moving items

Action Items

Priority	Action	Owner	Timeline
High	CR-2000 Inventory Reduction	Operations	Q2 2024
High	Demand Forecasting Implementation	IT/Ops	Q3 2024
Medium	Vendor Terms Negotiation	Procurement	Q2 2024
Medium	WIP Process Optimization	Operations	Q2-Q3 2024

Next Steps

- 1. Weekly monitoring of high-priority items
- 2. Bi-weekly cross-functional inventory review meetings
- 3. Monthly progress reports to executive team
- 4. Quarterly board updates on inventory optimization

Target for Q2 2024: - Turnover Ratio: 3.5x - DIO: 105 days - Working Capital Release: \$1.2M

This report contains confidential information and is intended for internal use only.